Final Exam for Algorithm and Complexity

已开始: 6月 16 17:55

测验说明

Final Exam for CS2308 Algorithm and Complexity

CS2308《算法与复杂性》期末考试

This exam consists of 2 parts and 5 questions, some of them includes several sub-questions. Please make sure you have checked all of the questions.

全卷共 2 部分，5 道题，其中部分题包含一些子问题。请仔细检查是否完成了所有的问题。

Part 1. Multiple Choice Questions (3.5'*10=35')

This part contains 10 multiple choice questions, each with 3.5 points. Please choose ONE correct answer for each question and directly click your answer in CANVAS.

(第一部分为单选题，共10道题目。每道题目请在四个选项中选择一个正确选项并直接在CANVAS中作答。)

问题 1

3.5 分

(MC1) Choose the correct time complexity of the following algorithm.

Algorithm 1: Toy Algorithm 1

Input: An integer n.

Output: Certain summation.

$sum \leftarrow 0$;

for $i \leftarrow 1$ ***to*** n **do**

  for $j \leftarrow 1$ ***to*** $\lfloor n/i \rfloor$ **do**

    $sum \leftarrow sum + i \times j$;

return sum;

- ○ $O(n)$
- ○ $O(n \log \log n)$
- ○ $O(n \log n)$

- ○ $O(n^2)$

问题 2

3.5 分

(MC2) Choose the stablest sorting algorithms among the following algorithms, *i.e.*, finding the sorting algorithm that has the best time complexity under the worst case.

- ○ Quick Sort
- ○ Merge Sort
- ○ Bubble Sort
- ○ Insertion Sort

问题 3

3.5 分

(MC3) Choose the correct correspondence between matroid and its Greedy-MAX algorithm.

Graphic Matroid M_G: Consider a (undirected) graph $G=(V,E)$, let $S=E$ and $\mathbf{C}$ the collection of all edge sets each of which induces an acyclic subgraph of G. Then $M_G=(S,\mathbf{C})$ is the graphic matroid.

Cographic Matroid M_G^*: Consider a (undirected) graph $G=(V,E)$, an subset $I\subseteq E$ is independent if the complementary subgraph $(V,E\backslash I)$ of G is connected. Let $S=E$ and $\mathbf{C}$ be the collection of such I, then $M_G^*=(S,\mathbf{C})$ is the cographic matroid.

- ○ Graphic Matroid - Prim Algorithm
- ○ Graphic Matroid - Kruskal Algorithm
- ○ Cographic Matroid - Kruskal Algorithm
- ○ Cographic Matroid - Prim Algorithm

问题 4

3.5 分

(MC4) About the components of sorting networks, which of the following statements is FALSE?

- ○ The depth of a Merger$[n]$ is the same as the depth of a Bitonic-Sorter$[n]$.
- ○ The depth of a Sorter$[n]$ is $O(\log n)$.
- ○ If the input to a Half-Cleaner is a bitonic sequence of 0's and 1's, then both the top half and the bottom half of the output are bitonic.
- ○ A Bitonic-Sorter of size n is composed of a Half-Cleaner of size n and two recursive Bitonic-Sorters of size $n/2$.

问题 5

3.5 分

(MC5) *Rod Cutting Problem*. Given a rod of total length N inches and a table of selling prices P_L for lengths $L = 1, 2, \ldots, M$. You are asked to find the maximum revenue R_N obtainable by cutting up the rod and selling the pieces. For example, based on the following table of prices, if we are to sell an 8-inch rod, the optimal solution is to cut it into two pieces of lengths 2 and 6, which produces revenue $R_8 = P_2 + P_6 = 5 + 17 = 22$. And if we are to sell a 3-inch rod, the best way is not to cut it at all.

Length L	1	2	3	4	5	6	7	8	9
Price P_L	1	5	8	9	10	17	17	20	23

Which of the following statements is FALSE?

- ○ If $N \leq M$, we have $R_N = \max\{P_N, \max_{1 \leq i < N}\{R_i + R_{N-i}\}\}$.
- ○ This problem can be solved by dynamic programming.
- ○ If $N > M$, we have $R_N = \max_{1 \leq i < N}\{R_i + R_{N-M}\}$.
- ○ The time complexity of this algorithm is $O(N^2)$.

问题 6

3.5 分

(MC6) *Coin Changing Problem*. Consider the problem of making change for n cents using the fewest number of coins. Assume that each coin's value is an integer. The coins of the lowest denomination is the cent.

(I) Suppose that the available coins are quarters (25 cents), dimes (10 cents), nickels (5 cents), and pennies (1 cent). The greedy algorithm always yields an optimal solution.

(II) Suppose that the available coins are in the denominations that are powers of c, that is, the denominations are $c^0, c^1, \ldots, c^k$ for some integers $c > 1$ and $k \geq 1$. The greedy algorithm always yields an optimal solution.

(III) Given any set of k different coin denominations which includes a penny (1 cent) so that there is a solution for every value of n, greedy algorithm always yields an optimal solution.

Which of the following statements is correct?

- ○ All of the three statements are correct.
- ○ Statement (II) is false.
- ○ Statement (I) is false.
- ○ Statement (III) is false.

问题 7

3.5 分

(MC7) For the given Linear Programming problem below, which one of the following is the correct dual form?

$$
\begin{array}{ll}
\text{minimize} & 2x_1 + x_3 + 4 \\
\text{subject to} & 3x_1 - x_2 + x_3 = 1 \\
 & x_1 + 2x_2 - 3x_3 \leq 2 \\
 & x_2 + 6x_3 \geq 5 \\
 & x_1 \geq 0, x_2 \geq 0
\end{array}
$$

- ○

$$\begin{array}{ll} \text{maximize} & y_1 + 2y_2 + 5y_3 + 4 \\ \text{subject to} & 3y_1 + y_2 \geq 2 \\ & -y_1 + 2y_2 + y_3 \geq 0 \\ & y_1 - 3y_2 + 6y_3 = 1 \\ & y_2 \geq 0, y_3 \leq 0 \end{array}$$

- ○ $$\begin{array}{ll} \text{maximize} & y_1 + 2y_2 + 5y_3 + 4 \\ \text{subject to} & 3y_1 + y_2 \geq 2 \\ & -y_1 + 2y_2 + y_3 \geq 0 \\ & y_1 - 3y_2 + 6y_3 = 1 \\ & y_2 \leq 0, y_3 \geq 0 \end{array}$$

- ○ $$\begin{array}{ll} \text{maximize} & y_1 + 2y_2 + 5y_3 - 4 \\ \text{subject to} & 3y_1 + y_2 \leq 2 \\ & -y_1 + 2y_2 + y_3 \leq 0 \\ & y_1 - 3y_2 + 6y_3 = 1 \\ & y_2 \geq 0, y_3 \leq 0 \end{array}$$

- ○ $$\begin{array}{ll} \text{maximize} & y_1 + 2y_2 + 5y_3 - 4 \\ \text{subject to} & 3y_1 + y_2 \geq 2 \\ & -y_1 + 2y_2 + y_3 \geq 0 \\ & y_1 - 3y_2 + 6y_3 = 1 \\ & y_2 \geq 0, y_3 \leq 0 \end{array}$$

问题 8

3.5 分

(MC8) When solving the problem All-Pairs Shortest Path by Floyd method, which one of the following iterations can give us the correct answer? (Assume that the graph is consists of n vertices, and define c as adjacency matrix of the shortest path between every two vertices)

- ○
  for $k \leftarrow 1$ ***to*** n **do**
      for $i \leftarrow 1$ ***to*** n **do**
          for $j \leftarrow 1$ ***to*** n **do**
              if $c_{ki} + c_{ij} < c_{kj}$ **then**
                  $c_{kj} = c_{ki} + c_{ij};$

- ○

```
for i ← 1 to n do
    for k ← 1 to n do
        for j ← 1 to n do
            if c_ki + c_ij < c_kj then
                c_kj = c_ki + c_ij;
```

○

```
for i ← 1 to n do
    for k ← 1 to n do
        for j ← 1 to n do
            if c_ik + c_kj < c_ij then
                c_ij = c_ik + c_kj;
```

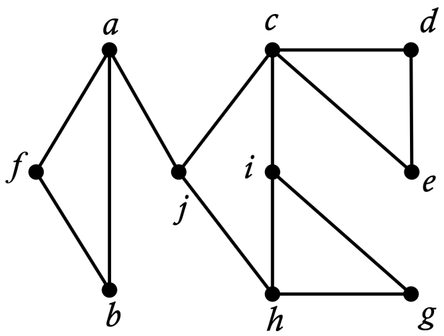
○

```
for i ← 1 to n do
    for j ← 1 to n do
        for k ← 1 to n do
            if c_ik + c_kj < c_ij then
                c_ij = c_ik + c_kj;
```

问题 9

3.5 分

(MC9) Given the graph G shown as follows, assume we start from vertex a and use alphabetical order to explore G, then choose the correct correspondence between the exploration method and the exploration order.



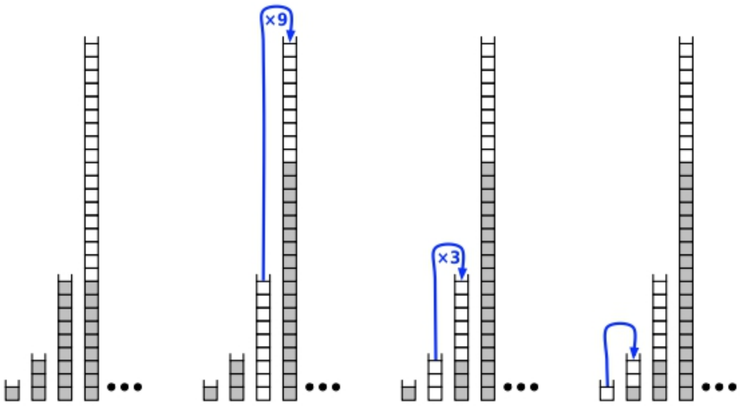
- ○ Breadth-First-Search: a, b, f, j, c, h, d, e, g, i
- ○ Depth-First-Search: a, b, f, j, c, d, e, h, g, i
- ○ Depth-First-Search: a, b, f, j, c, d, e, i, h, g

○ Breadth-First-Search: a, b, f, j, c, d, e, g, h, i

问题 10

3.5 分

(MC10) A multistack consists of an infinite series of ordinary stacks $S_0, S_1, \cdots, S_n, \cdots$, where the i-th stack S_i has the size of 3^i. For push operation of the multistack, we first intend to push the element onto S_0. If S_i is full, then we intend to pop all elements off S_i and push them onto S_{i+1} to make room. (If S_{i+1} is already full, then we recursively move all its members to S_{i+2}.) The following figure shows an illustrative example of making room for one new element in multistack. For every stack S_i, a push/pop operation takes $O(1)$ time. In the best, average and worst case, how long does it take to push a new element onto a multistack containing n elements.



Making room for one new element in a multistack.

○ $O(\log n)$, $O\left(\log^2 n\right)$, $O(n)$

○ $O(1)$, $O\left(\log^2 n\right)$, $O(n)$

○ $O(1)$, $O(\log n)$, $O(n)$

○ $O(1)$, $O(n)$, $O(3^n)$

Part 2. Questions and Answers (65')

This part contains 4 questions, some of them may contain multiple sub-questions. Please write down your solution on answer sheet, and upload the scanned version through CANVAS.

(第二部分为问答题，包含4道大题，每道大题可能有多道小题。请将你的答案写在答题纸上并拍照上传。)

(QA1, Divide-and-Conquer, 15')

(a) Write down the Master's Theorem that we have learned in the class. (**Hint**: If $T(n) = aT(\lceil n/b \rceil) + O(n^d)$ for some constants $a > 0, b > 1, d \geq 0$, then ...)

(b) Consider the following recurrence.

$$T(n) = 2T\left(\left\lceil \frac{n}{2} \right\rceil\right) + O(n \log n).$$

Can it be solved via the Master's Theorem in problem (a), why or why not? If true, write down the time complexity directly; otherwise, please also provide derivations.

(c) A perfect permutation Π is a permutation of integers $1, 2, \cdots, n$ which satisfies the following condition: there is no index k with $1 \leq i < k < j \leq n$ where $2\Pi_k = \Pi_i + \Pi_j$. Given an integer n, please design an efficient algorithm in pseudo-code to generate a perfect permutation Π of length n.

(QA2, Greedy Strategy & Dynamic Programming, 18')

Little Gyro loves to watch TV series, however, the TV series on the market are so dazzling that Little Gyro usually doesn't know which episode should be selected to watch. Please help him design algorithms to solve the following problems:

(a) Suppose that Little Gyro has one TV set and can only watch one TV series at the same time. In order to get the greatest satisfaction, Little Gyro decides to watch as more TV series as possible. Given n TV series and for the i-th TV play define its start time t_i^s and end time t_i^e, please briefly describe your algorithm and analyze its time complexity and space complexity.

(b) Suppose that Little Gyro has a laptop and can watch TV series in a certain period of time T. In order to get the greatest satisfaction, Little Gyro decides to watch the TV series which could gain more happiness. Given n TV series, a certain period of time T

and for the i-th TV play define its happiness h_i and time cost t_i, please briefly describe your algorithm and analyze its time complexity and space complexity.

(QA3, Network Flow, 17')

Let $G = (V, E)$ be a connected directed graph. Set k starting vertices $S = \{s_1, s_2, \ldots, s_k\} \in V$ and k terminal vertices in $T = \{t_1, t_2, \ldots, t_k\} \in V$. Your goal is to find more possible paths from the starting vertices s_i to any terminal vertices b_j such that no two paths share any edges, and no two paths end in the same vertex b_j.

(a) Design an algorithm to find more possible paths $s_i \rightarrow t_j$ that start and end at different vertices and such that they do not share any edges.

(b) Modify your algorithm to find more possible paths $s_i \rightarrow t_j$, that start and end in different vertices and such that they share neither vertices nor edges.

(QA4, NP and Polynomial-Time Reduction, 15')

PARTITION. Given n non-negative integers $a_1, a_2, \cdots, a_n$, does there exist a subset S of $\{1, 2, \cdots, n\}$ such that

$\sum_{i \in S} a_i = \sum_{j \in \{1,2,\cdots,n\} \setminus S} a_j$.

(a) Write down the certificate and certifier of the *PARTITION* problem, and show that it is an NP problem.

(b) Show that *PARTITION* is NP-complete. (**Hint**: recall the *SUBSET-SUM* problem.)

(c) Suppose n is even and we require that the subset S contains exactly $\frac{n}{2}$ elements, is the *PARTITION* problem under these constraints still NP-complete? Justify your answer.

问题 11

65 分

Upload The Answer Sheet（问答题部分答题纸提交处）：

Please upload your answer sheets using .pdf, .png, .jpeg or .jpg files. Make sure you've uploaded each page and name your file correctly.

请于此处上传答题纸（问答题答案），注意每页答题纸必须有姓名和学号，请写清题号，对上传文档规范文件命名，命名格式为“Algorithm.pdf”。如果有多个文档，请用“Algorithm01.pdf”等命名上传。

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